Palestine Violence Archive

Dashboard & Pipeline User Guide

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This guide explains how to use the Streamlit pipeline dashboard,
what each page does, how research assistants should review batches,
and how to export data for statistical analysis.

1. Project Overview

The Palestine Violence Archive is a research pipeline built for a political science lab at UC San Diego. It automatically collects news articles about human rights incidents affecting Palestinians in the West Bank from the WAFA news agency, classifies each incident across 10 violence labels using machine learning, routes uncertain cases to human reviewers, and continuously retrains the model as reviewed labels accumulate.

The goal is to produce a clean, labeled dataset of incidents suitable for statistical analysis in SPSS or R and for export to a permanent research database in support of peer-reviewed publication.

2. Google Sheets Structure

The pipeline uses three tabs in one shared Google Spreadsheet:

Reports - Raw scraped articles (input layer). Written by the scraper. Contains: report_id, url, title, body_text, published_date, language, location_raw, actors_raw.

Review - Model predictions awaiting human verification (working layer). Written by the pipeline after each prediction run. RAs fill in the human_* columns here and set review_status = reviewed.

Incidents - Finalized, clean incident records (publication layer). Written by the Transfer to Incidents step after human review is complete. Contains only facts and labels - no model scores or pipeline metadata. This tab mirrors what a proper database table will look like, making future migration straightforward.

3. Pipeline Flow

The pipeline runs in seven steps:

Step 1 - Ingest - Scrapes new WAFA articles into the Reports tab. Duplicate URLs are skipped automatically.

Step 2 - Export - Reads unprocessed rows from Reports and saves them as a timestamped CSV in data/incoming/.

Step 3 - Predict - Runs the baseline or DeBERTa model. Produces a compact review sheet in data/review_exports/ and pushes it to the Review tab.

Step 4 - Human Review - RAs open the Review tab, fill in human_* columns (1/0), and set review_status = reviewed. Double-review rows (5% random sample) are filled by a second RA for inter-rater reliability tracking.

Step 5 - Merge - Reviewed labels are merged into master_reviewed_dataset.csv for model retraining.

Step 6 - Transfer to Incidents - Clean confirmed records are written to the Incidents tab, stripping all pipeline columns. Run after each merge.

Step 7 - Retrain - Once enough labeled data accumulates, the baseline model is rebuilt. DeBERTa can be fine-tuned locally with a GPU.

4. Dashboard Pages

The left sidebar lists six pages. Each is described below.

4a. Dashboard

The landing page gives a live snapshot of the pipeline state.

Incoming batch size - Articles in data/incoming/ not yet reviewed.

Review queue - Rows in the latest review sheet with review_status not reviewed.

Model metrics - Per-label F1, precision, and recall from the last baseline training run (read from baseline_meta.json).

IRR panel - Cohen's Kappa inter-rater reliability for double-reviewed rows. A macro Kappa above 0.80 indicates strong RA agreement. Click 'Compute IRR' to refresh after RAs finish a batch.

4b. Run Pipeline

The main control panel for Steps 1-3 (ingest, export, predict).

Model - 'baseline' runs on CPU and works on Streamlit Cloud. 'deberta' is more accurate but requires a GPU locally.

Skip ingest - Use existing Sheets data without scraping new articles.

Re-export all - Forces re-export ignoring the already-exported tracker.

Push to Sheets - After predicting, uploads the review sheet to the Review tab so RAs can work collaboratively online.

Dry run - Prints what each step would do without writing any files.

Deduplicate - Removes near-duplicate articles before prediction.

Max rows - Caps the batch. Recommended: 100-150 for manageable reviews.

Log output - Streams live and persists after the run so you can scroll it.

4c. Review Access

Access to review files and the Google Sheets Review tab.

Google Sheets link - Opens the spreadsheet. Direct RAs to the 'Review' tab.

Latest review file - Preview of the most recent review_compact_*.csv.

Latest predictions - Raw inference audit trail with all confidence scores.

RA workflow reminder - Step-by-step instructions shown on-screen.

4d. Export Analysis

Exports the cleaned incident database for the professor (SPSS/R).

Years - Which years to include (default: 2023 2024 2025). Requires data/processed/oppression_{year}.csv.

Output directory - Where to write files (default: analysis/).

Skip SPSS export - Check this if pyreadstat is not installed.

Exported files - .sav (SPSS/JASP), .csv (R/Excel), and codebook_variables.csv. The .sav has full variable labels and 0=No / 1=Yes value labels for all binary columns.

R analysis script - analysis/descriptive_analysis.R produces 11 summary tables and 7 figures. Run with: Rscript analysis/descriptive_analysis.R

4e. Retrain (Admin only)

Rebuilds the model and manages the Incidents tab.

Training data stats - Current row count of master_reviewed_dataset.csv.

Checkpoint system - Auto-creates a timestamped backup before any destructive action. Manually create or restore checkpoints at any time.

Merge reviewed labels - Pulls reviewed rows from the Sheets Review tab and merges human labels into the master training CSV. Auto-checkpoints before merging.

Transfer to Incidents - Copies confirmed records (review_status = reviewed) from the Review tab into the clean Incidents tab, stripping all pipeline columns. Run after every merge to keep the Incidents tab current for the database.

Retrain baseline - Rebuilds the TF-IDF + logistic regression model. 1-5 minutes on CPU.

Retrain DeBERTa - Fine-tunes the transformer. Requires GPU; plan 1-3 hours.

4f. Documentation

This page. Contains the full inline user guide and a Download PDF Guide button that generates and downloads this document.

5. RA Review Workflow

Research assistants follow this process for each batch:

Step 1 - Open Google Sheets and go to the Review tab (or download the CSV from data/review_exports/).

Step 2 - For each row, read the description and fill the 10 human_* columns with 1 (yes) or 0 (no). Leave blank if uncertain; add reviewer_notes.

Step 3 - For rows marked double_review = TRUE, a second RA independently fills the human2_* columns without seeing the first RA's answers. This measures inter-rater reliability.

Step 4 - Set review_status = reviewed for each completed row.

Step 5 - Notify the pipeline operator who will run Merge then Transfer to Incidents.

6. Incident Labels

The model classifies each incident into 10 binary labels (1 = present):

Raid - Organized entry by Israeli forces into a Palestinian area.

Arrest/Detention - Apprehension or detention by Israeli forces or the PA.

Physical Assault - Shooting, beating, tear gas, stun grenades, dogs.

Coercive Actions - Intimidation or harassment without direct assault.

Restriction of Freedoms - Curfews, checkpoints, roadblocks.

Religious Encroachment - Incursion into or desecration of religious sites.

Harm to Property - Vandalism or destruction of Palestinian property.

Dispossession - Seizure or theft of land, livestock, crops, or assets.

Protest - Palestinian organized demonstration or civil resistance.

Multi-Community Incident - Coordinated action across multiple communities.

7. Models

Baseline (TF-IDF + Logistic Regression)
Location: modeling/saved_models/baseline_rank_based_fixed/
Runs on CPU. Used by the Streamlit Cloud dashboard.
Rank-based thresholding: predicts top-k articles per label where k matches the training prevalence rate.

DeBERTa v3 (Transformer)
Location: modeling/saved_models/deberta_v3_multilabel/
Weights: HuggingFace Hub (jalva182/palestine-violence-deberta)
Requires GPU. Install with: pip install -r requirements-gpu.txt

Both models output a confidence score (0-1) per label. Rows within 0.10 of the threshold are flagged needs_review = TRUE.

8. Streamlit Cloud Setup

1. Connect the repo at share.streamlit.io. Set main file: streamlit_app/app.py

2. Add Google credentials in app Settings -> Secrets:

```
[gcp_service_account]
type = "service_account"
project_id = "your-project-id"
private_key = "-----BEGIN PRIVATE KEY-----\n...\n-----END PRIVATE KEY-----\n"
client_email = "...@....iam.gserviceaccount.com"
... (all other fields from service_account.json)
```

3. DeBERTa is not available on Streamlit Cloud (torch too large). Use the baseline model on Cloud; run DeBERTa locally.

4. After updating secrets or pushing code changes, use three-dot menu -> Reboot app to force a full restart.

9. File Locations

```
data/incoming/        - Raw batches exported from Google Sheets
  data/review_exports/   - Model predictions + compact review sheets
  data/reviewed/        - Completed RA review files (local copies)
  data/training/        - Master labeled dataset (used for retraining)
  data/processed/       - Normalized yearly CSVs (2023, 2024, 2025)
  analysis/             - SPSS .sav, R CSV, codebook, R analysis script
  modeling/saved_models/ - Baseline and DeBERTa model artifacts
  modeling/backups/      - Timestamped model checkpoints
  logs/                 - Pipeline run logs (pipeline_YYYYMMDD.log)
  config/               - Label codebook YAML
  docs/                 - PDF user guide and generator script
  requirements.txt       - Streamlit Cloud dependencies (no torch)
  requirements-gpu.txt   - Local DeBERTa dependencies (torch + transformers)
```

10. Common Issues

100% of rows flagged for review

Normal for the first batch before the model is retrained on your data. Confidence improves as reviewed batches are merged and the model is retrained.

No articles scraped (0 new reports)

Check the log for robots.txt warnings. Use Skip Ingest and re-run.

service_account.json not found

Add credentials to Streamlit Cloud Secrets under [gcp_service_account].

DeBERTa weights not found

Download from HuggingFace Hub into modeling/saved_models/deberta_v3_multilabel/deberta_v3_best/

Install GPU deps first: pip install -r requirements-gpu.txt

pyreadstat not installed

pip install pyreadstat or check 'Skip SPSS export' on the Export page.

Transfer to Incidents shows 0 rows

No rows have review_status = reviewed yet. RAs must complete their review first.